05. Angles & 2D Shapes

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Classifying an angle:

What type of angle is 115°?

Answer: It is between 90° and 180°, so it is **obtuse**.

Example 2 — Angles on a straight line:

Two angles on a straight line are 65° and x°. Work out x.

Answer: Angles on a straight line always add to 180°. x = 180 − 65 = **115°**.

Example 3 — Angles around a point:

Three angles meet at a single point: 90°, 120° and x°. Work out x.

Answer: Angles around a point always add to 360°. x = 360 − 90 − 120 = **150°**.

Example 4 — Interior angles of a triangle:

A triangle has angles of 50° and 65°. Work out the third angle.

Answer: The three interior angles of any triangle always add to 180°. Third angle = 180 − 50 − 65 = **65°**.

Example 5 — Isosceles triangle, multi-step:

An isosceles triangle has two equal angles of 48° each. What is the third angle?

Answer: Add the two equal angles first: 48 + 48 = 96°. Then subtract from 180°: 180 − 96 = **84°**.

Foundation

15 questions

1. What type of angle is 25°?

2. What type of angle is 95°?

3. What type of angle is exactly 90°?

4. What type of angle is exactly 180°?

5. Which of these is an acute angle? A) 95° B) 45° C) 180° D) 270°

6. How many sides does a hexagon have?

7. How many sides does an octagon have?

8. Which shape has 3 sides? A) square B) triangle C) pentagon D) hexagon

9. How many lines of symmetry does a square have?

10. How many lines of symmetry does an equilateral triangle have?

11. Select ALL of these angles that are OBTUSE (between 90° and 180°): A) 45° B) 170° C) 90° D) 120° E) 200°

12. What is the name for an angle greater than 180° but less than 360°?

13. How many right angles does a rectangle have?

14. Which shape has 4 equal sides and 4 right angles? A) rectangle B) square C) rhombus D) kite

15. How many sides does a pentagon have?

Standard

20 questions

1. On a straight line, two angles are 72° and x°. Find x.

2. Three angles meet at a point: 95°, 140° and x°. Find x.

3. A triangle has angles of 55° and 60°. Find the third angle.

4. What type of triangle has all three sides different lengths?

5. Which triangle has exactly two sides equal? A) scalene B) isosceles C) equilateral D) right-angled

6. Angles on a straight line always add up to how many degrees?

7. Angles around a point always add up to how many degrees?

8. What is the sum of the interior angles of a triangle?

9. What is the sum of the interior angles of a quadrilateral?

10. Select ALL of the following that are quadrilaterals (4-sided shapes): A) square B) pentagon C) rhombus D) rectangle E) triangle

11. A right angle is split into two parts. One part is 37°. What is the other part?

12. Two straight lines cross. One of the four angles formed is 48°. What size is the angle directly opposite it (vertically opposite)?

13. In an isosceles triangle, two equal angles are each 70°. What is the third angle?

14. A quadrilateral has three angles of 85°, 95° and 100°. What is the fourth angle?

15. How many lines of symmetry does a regular hexagon have?

16. Which of these shapes has no straight sides? A) circle B) square C) triangle D) hexagon

17. A triangle has one right angle, and the other two angles are equal. What size is each of the other two?

18. Select ALL of the following statements that are TRUE: A) A square is a type of rectangle B) All rectangles are squares C) Every rhombus has 4 equal sides D) A pentagon has 5 sides

19. Three equal angles meet at a point. What size is each one?

20. A straight line is divided into two equal angles. What size is each one?

Hard

5 questions

1. Two angles on a straight line are in the ratio 2:3. What are the two angles?

2. Three angles around a point are in the ratio 1:2:3. Find the size of the largest angle.

3. A quadrilateral's four angles are in the ratio 2:3:4:3. Find the size of the largest angle.

4. Two straight lines cross. One of the angles formed is 64°. What is the size of the angle on the straight line next to it (not the one directly opposite)?

5. A triangle has angles 42°, 58° and x°. Select ALL the statements below that are TRUE: A) x = 80° B) The triangle is scalene (all three angles are different) C) The triangle contains an obtuse angle D) The triangle contains a right angle

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